

MATLAB Assignment 6

In MATLAB Assignment 5, we focused on running the code and generating results without delving too deeply into the underlying functions. However, after completing the assignment, we gained a better understanding of the code structure and the purpose of the individual functions.

Now, we aim to take a closer look at the main components of the code, not by examining every line in detail, but by identifying and understanding the key pieces. Our goal is to interpret how each part contributes to the overall functionality and to connect these components with the theoretical framework of radial basis function (RBF) interpolation. We will also relate our observations to the concepts presented in Paper 1.

RBF_interp_students.m

Step 1:

`get_cm` code is generating the new data (i.e., coordinates of green dots and the corresponding signal). Let's explore how it is doing that.

Talk with your neighbor about what's happening until line 13.

There are zoning in on the neighboring q-balls

`2*irad+1, 2*irad+1, 2*ngae` explain what this is about.

These are the radii for the voxel of red points in 3D matrix.

`nbd=4;` What does this 4 mean in relation to paper 1?

This is signifying the 4 nearest red signal values neighbors.

`for im=1:ntgt,` what is this for loop running over?

Loops over all blue directions for signal approximation.

`for ik=1:2*ngae,` what is this for loop running over?

`dst(ik)=norm(udirt(im,:)-erdirt(ik,:));` What is this line about?

The Euclidean distance between target direction and current known direction

`%`

`end`

`[~,ind]=sort(dst);` What is this line about?

Save index positions of your nearest neighbors after sorting from closest to farthest.

`idst=zeros(nbd,1);` What is this line about? At the end of the loop, what does `idst` contain?

`for ii=1:nbd`

`idst(ii)=ind(ii);`

`end`

`val=zeros(3,3,nbd);`

`vec=zeros(nbd,3);`

`for ii=1:nbd`

`val(:, :, ii)=squeeze(pxrho(:, :, idst(ii)));` what does `val` contain?

`vec(ii, :)=erdirt(idst(ii), :);` what does `vec` contain?

Stores 3D coordinates for the nearest points equation (5)

`end`

```

for it=1:4

    ith=circshift([1:nbd]',it-1)';
    tri_val=val(:, :, ith(1:3));
    tri_vec=vec(ith(1:3),:);

    valcorr=zeros(3,1);
    tmp=corrcoef(squeeze(tri_val(:, :, 1)),squeeze(tri_val(:, :, 2)));
    valcorr(1)=(tmp(1,2));
    tmp=corrcoef(squeeze(tri_val(:, :, 2)),squeeze(tri_val(:, :, 3)));
    valcorr(2)=(tmp(1,2));
    tmp=corrcoef(squeeze(tri_val(:, :, 3)),squeeze(tri_val(:, :, 1)));
    valcorr(3)=(tmp(1,2));

```

What does valcorr contain?



```

if sum(valcorr > cut)==3 Explain this if loop, refer to paper 1.

```

Tries to avoid noisy interpolation through cross-checking triangular correlation coefficients to if it exceeds cut variable value.

```

w1= valcorr(1)+valcorr(3);
w2= valcorr(1)+valcorr(2);
w3= valcorr(2)+valcorr(3);
w=w1+w2+w3;

cmdirt(cmdix,:)=(w1*tri_vec(1,:)+w2*tri_vec(2,:)+w3*tri_vec(3,:))./w;
cmval(cmdix)=(w1*tri_val(2,2,1)+w2*tri_val(2,2,2)+w3*tri_val(2,2,3))./w;
cmdix=cmdix+1;

```

```

end

```

```

end

```

```

end

```

```

cmdirt=cmdirt(1:cmdix-1,:);
cmval=cmval(1:cmdix-1)';
[cmdirt,cmdix]=unique(cmdirt,'rows');
cmval=cmval(cmdix);

```

Step 2:

`K_cm = convhulln(rgb_directions);` what does this in-built MATLAB code do?

This computes the convex hull for all combination directions.

Add a code line to visualize this convex hull using other inbuilt MATLAB functions.

```

trisurf(K_cm, rgb_directions(:,1), rgb_directions(:,2), rgb_directions(:,3));

```

Step 3:

Briefly explain what **LOOCV3Dmin** code is doing in just a couple of sentences. Refer to paper 1 as necessary.

This function optimizes the RBF's kernel parameter through leave-one-out cv.

This regularization parameter is key to finding the weights coefficients (9)

Step 4:

Briefly explain what **RBFInterpolation** code is doing in just a couple of sentences. Refer to paper 1 as necessary.

Approximates solution to the linear system of equation (8) using Gaussian RBF with lambda regularization.

Briefly explain what **DistanceMatrix** code is doing in just a couple of sentences. Refer to paper 1 as necessary.

Computes Euclidean distance for the RBF approximations.

Step 5:

Estimate the blue signal for a selected brain location (ii,ij) using RBF interpolation using Gaussian radial basis function as well as inverse multiquadric function.

Compute the L2 error for the blue signal for each case.

Which gives the lowest L2 error for the chosen brain location.

$$b = \gamma^2 \delta^2 \left(\Delta - \frac{\delta}{3} \right) \|\mathbf{g}\|^2.$$

The EAP is assumed to be radially symmetric, i.e. $P(\mathbf{R}) = P(-\mathbf{R})$. Diffusion in the x -, y -, and z -directions can be measured by applying the gradient in the x -, y -, and z -directions, respectively.

A typical HARDI signal is obtained from spherical sampling, and so a spherical basis is an appropriate representation for a raw HARDI signal. The HARDI requires a finer angular resolution sampling comparing to the conventional DTI sampling scheme. The HARDI methods estimate ODF in multiple shell or single shell sampling data. In spite of principal advantage of q -space approaches, the main disadvantage is that the HARDI technique requires a large number of measurements to encode q -space and thus it takes a long acquisition time, restricting its use in the clinical implementation.

2.2. Prediction of diffusion gradient direction using a local diffusion pattern map

For a given voxel coordinate $\mathbf{r}$ in an imaging region and a set $\mathcal{D}_N = \{\mathbf{g}_1, \dots, \mathbf{g}_N\}$ of DG directions, let $S(\mathbf{g}_i) = S(\mathbf{g}_i, \mathbf{r})$, $i = 1, \dots, N$, be the measured diffusion MR signals for the DG directions $\mathcal{D}_N$ and a fixed b -value.

In this paper, we aim to generate appropriate diffusion MR signals on a finer set $\mathcal{D}_M$ of DG directions including the DG directions $\mathcal{D}_N$. We denote the elements of $\mathcal{D}_M$ as $\mathcal{D}_M = \{\mathbf{g}_1, \dots, \mathbf{g}_N, \mathbf{v}_1, \dots, \mathbf{v}_{M-N}\}$. Instead of a simple approximation method, we shall generate diffusion MR signals, $S(\mathbf{v}_j)$, $j = 1, \dots, M - N$, by taking into account the similarity of the measured MR signals $S(\mathbf{g}_i)$ in neighborhood DG directions. For a direction $\mathbf{v}_j \in \mathcal{D}_M \setminus \mathcal{D}_N$, let $\mathcal{N}_{\mathbf{v}_j}^K = \{\mathbf{g}_k \in \mathcal{D}_N : k = 1, \dots, K\}$ be the set of K directions $\mathbf{g}_k$ close to $\mathbf{v}_j$.

A basic motivation is that local diffusion pattern similarities in the neighborhood $\mathcal{N}_{\mathbf{v}_j}^K$ are related to the diffusion signals in the convex hull, $\text{Conv}(\{\mathbf{g}_1, \dots, \mathbf{g}_K\}) = \{\sum_{k=1}^K w_k \mathbf{g}_k : \sum_{k=1}^K w_k = 1, w_k \geq 0\}$:

$$\begin{aligned} p_n(S(\mathbf{g}_{i_n})) &\approx p_n(S(\mathbf{g}_{i_m})) \text{ for } m, n = 1, \dots, K \text{ implies} \\ S(\mathbf{u}_c) &:= S\left(\sum_{k=1}^K w_k \mathbf{g}_k\right) \approx \sum_{k=1}^K w_k S(\mathbf{g}_k), \end{aligned} \quad (1)$$

where $p_n(S(\mathbf{g}_{i_n}))$ denotes a local diffusion pattern map of the diffusion signal $S(\mathbf{g}_{i_n})$ and $\mathbf{u}_c \in \text{Conv}(\{\mathbf{g}_1, \dots, \mathbf{g}_K\})$. Depending on the similarity between the local diffusion patterns in $\mathcal{N}_{\mathbf{v}_j}^K$, we determine a new appropriate direction

$$\mathbf{u}_c = \sum_{k=1}^K w_k \mathbf{g}_k \quad (2)$$

in $\text{Conv}(\{\mathbf{g}_1, \dots, \mathbf{g}_K\})$, which is synthesized by the measured directions in the neighborhood $\mathcal{N}_{\mathbf{v}_j}^K \subset \mathcal{D}_N$. In this paper, we take a local diffusion pattern map, p_n in (1), at a spatial voxel $\mathbf{r} = (x, y, z)$ between two diffusion MR signals $S(\mathbf{g}_{i_n})$ and $S(\mathbf{g}_{i_m})$ as a correlation coefficient map using a $3 \times 3 \times 3$ local window $\omega_r = \{(x + l_x, y + l_y, z + l_z) : l_x, l_y, l_z = -1, 0, 1\}$ centered at $\mathbf{r}$. Let $S_{j_n} = \{S(\mathbf{g}_{i_n}, (x + l_x, y + l_y, z + l_z)) : l_x, l_y, l_z = -1, 0, 1\}$ be the set of diffusion MR signals around $S(\mathbf{g}_{i_n})$ in the local window ω_r . The correlation coefficient $CC(S_{j_n}, S_{j_m})$ between two set of neighboring diffusion MR signals S_{j_n} and S_{j_m} is defined as

$$CC(S_{j_n}, S_{j_m}) = \frac{\text{Cov}(S_{j_n}, S_{j_m})}{\sqrt{\text{Cov}(S_{j_n}, S_{j_n}) \text{Cov}(S_{j_m}, S_{j_m})}} \quad (3)$$

where

$$\begin{aligned} \text{Cov}(S_{j_n}, S_{j_m}) &= \frac{1}{27} \sum_{l_x, l_y, l_z = -1}^1 \left(S(\mathbf{g}_{i_n}, (x + l_x, y + l_y, z + l_z)) - \bar{S}_{j_n} \right) \times \\ &\quad \left(S(\mathbf{g}_{i_m}, (x + l_x, y + l_y, z + l_z)) - \bar{S}_{j_m} \right) \end{aligned}$$

Here, $\bar{S}_{j_n}$ is the average of MR signals in S_{j_n} .

To determine the weighting factors w_k , $k = 1, \dots, K$ in (2), we use the correlation coefficient $CC(S_{j_n}, S_{j_m})$ as a local diffusion pattern map for $S(\mathbf{g}_{i_n})$ and $S(\mathbf{g}_{i_m})$. Define

$$\mathbf{A} = \begin{bmatrix} a_{11} & \dots & a_{1K} \\ \vdots & \ddots & \vdots \\ a_{K1} & \dots & a_{KK} \end{bmatrix}$$

where

$$a_{mn} = \begin{cases} CC(S_{j_n}, S_{j_n}), & m \neq n \\ 0, & m = n. \end{cases}$$

For the measured DG direction $\mathbf{g}_{i_j}$, $j \in \{1, \dots, K\}$, we set a similarity weight as

$$w_k = \sum_{n=1}^K a_{kn} / \|\mathbf{A}\|$$

where $\|\mathbf{A}\| = \sum_{m=1}^K \sum_{n=1}^K a_{mn}$. Using these weights, we determine a new predicted DG direction

$$\mathbf{u}_c = \sum_{k=1}^K w_k \mathbf{g}_{i_k} = \sum_{k=1}^K \sum_{n=1}^K a_{kn} / \|\mathbf{A}\| \mathbf{g}_{i_k} \in \text{Conv}(\mathcal{N}_{\mathbf{v}_j}^K). \quad (4)$$

This direction $\mathbf{u}_c$ satisfies a balancing condition

$$\frac{1}{\|\mathbf{A}\|} \sum_{n=1}^K \sum_{k=1}^K a_{kn} (\mathbf{g}_{i_n} - \mathbf{u}_c) = 0.$$

The proposed method to predict a DG direction uses the diffusion similarity and determines the DG direction using a linear combination of the surrounding DG directions.

To specify the proposed method and to select more predicted directions, we first select the four nearest directions $\{\mathbf{g}_{i_k} \in \mathcal{D}_N : k = 1, \dots, 4\}$ around $\mathbf{v}_j \in \mathcal{D}_M \setminus \mathcal{D}_N$ and then select the four triangles as shown in Fig. 1(a). One of the selected triangles is denoted by $\mathcal{N}_{\mathbf{v}_j}^{3,1}$.

The correlation coefficients in the local neighborhood $\mathcal{N}_{\mathbf{v}_j}^{3,1}$ determine a location as an intermediate DG direction. For the local neighborhood $\mathcal{N}_{\mathbf{v}_j}^{3,1} = \{\mathbf{g}_{i_1}, \mathbf{g}_{i_2}, \mathbf{g}_{i_3}\}$, we predict an appropriate DG direction, $\mathbf{u}_{c_1}$, only when the correlation coefficient provides a positive relation; $CC(S_{j_1}, S_{j_2}) > 0$, $CC(S_{j_2}, S_{j_3}) > 0$, and $CC(S_{j_3}, S_{j_1}) > 0$. In that case, a new predicted DG direction $\mathbf{u}_{c_1}$ is determined as follows:

$$\mathbf{u}_{c_1} = \sum_{k=1}^3 w_k \mathbf{g}_{i_k}$$

where

$$w_1 = \frac{CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3})}{2(CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3}) + CC(S_{j_3}, S_{j_1}))},$$

$$w_2 = \frac{CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3})}{2(CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3}) + CC(S_{j_3}, S_{j_1}))},$$

$$\text{and } w_3 = \frac{CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3})}{2(CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3}) + CC(S_{j_3}, S_{j_1}))}.$$

Repeating this procedure for all triangles $\mathcal{N}_{\mathbf{v}_j}^{3,1}$ in $\{\mathbf{g}_{i_k} \in \mathcal{D}_N : k = 1, \dots, 4\}$, we can include up to four predicted positions. As an example, Fig. 1(a) represents the selected DG directions $\{\mathbf{g}_{i_k} \in \mathcal{D}_N : k = 1, \dots, 4\}$.

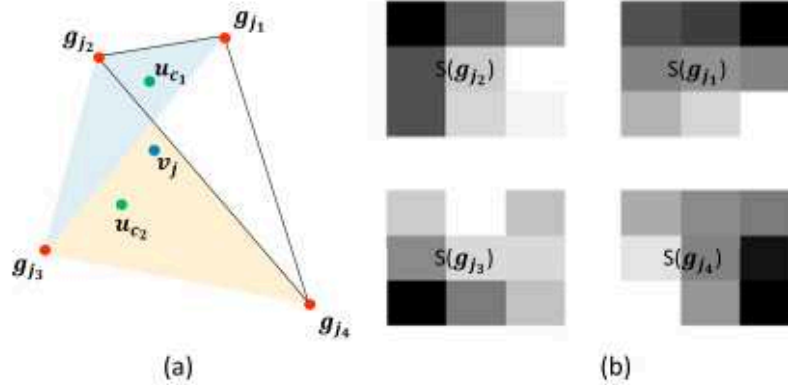


Fig. 1. Diagram for predicting directions in $\text{Conv}(\{g_{j1}, g_{j2}, g_{j3}, g_{j4}\})$. (a) Selected DG directions $\{g_{j1}, g_{j2}, g_{j3}, g_{j4}\}$ and determined predicted directions u_{c1} and u_{c2} within triangles $\mathcal{N}_{v_j}^{3,1} = \{g_{j1}, g_{j2}, g_{j3}\}$ and $\mathcal{N}_{v_j}^{3,2} = \{g_{j2}, g_{j3}, g_{j4}\}$, respectively. (b) Intensity of $S(g_{jk}, (x + l_x, y + l_y, z + l_z))$, $l_x, l_y = -1, 0, 1$, $l_z = 0$, $k = 1, \dots, 4$, in a $3 \times 3 \times 3$ local window.

and determined predicted positions u_{c1} and u_{c2} within triangles $\mathcal{N}_{v_j}^{3,1} = \{g_{j1}, g_{j2}, g_{j3}\}$ and $\mathcal{N}_{v_j}^{3,2} = \{g_{j2}, g_{j3}, g_{j4}\}$, respectively. Fig. 1(b) shows a portion of MR signals in a $3 \times 3 \times 3$ local window in given DG directions. The diffusion patterns are used to calculate the weighting factors w_k within the triangle neighborhoods $\mathcal{N}_{v_j}^3$.

2.3. Synthesized MR diffusion signal at a predicted gradient direction

The weighted combination of the correlation coefficients $CC(S_{g_{j1}}, S_{g_{j2}})$ between two set of neighboring diffusion MR signals $S_{g_{j1}}$ and $S_{g_{j2}}$ determines the location u_c which is a convex combination of DG directions in $\mathcal{N}_{v_j}^3$. Since the weighting factors w_k for the predicted DG direction, $u_c = \sum_{k=1}^K w_k g_{jk}$ in (4), is determined with respect to the diffusion similarity, using the same weighting factors, we generate a synthesized MR diffusion signal at u_c as

$$S(u_c) = S\left(\sum_{k=1}^K w_k g_{jk}\right) = \sum_{k=1}^K w_k S(g_{jk}) \quad (5)$$

For the unmeasured true MR diffusion signal, $S^r(u_c)$, at $u_c \in \text{Conv}(\mathcal{N}_{v_j}^3)$ and the estimated $S(u_c)$, the error between $S^r(u_c)$ and $S(u_c)$ can be decomposed into the following form:

$$\begin{aligned} |S^r(u_c) - S(u_c)| &= \left| S^r(u_c) - \sum_{k=1}^K w_k S^r(g_{jk}) + \sum_{k=1}^K w_k (S^r(g_{jk}) - S(g_{jk})) \right| \\ &\leq \left| S^r(u_c) - \sum_{k=1}^K w_k S^r(g_{jk}) \right| + \left| \sum_{k=1}^K w_k (S^r(g_{jk}) - S(g_{jk})) \right| \end{aligned}$$

The first term on the right side of the last expression shows the reliability error from the prediction model. We can express the reliability error in the following form:

$$\left| S^r(u_c) - \sum_{k=1}^K w_k S^r(g_{jk}) \right| = \left| S^r\left(\sum_{k=1}^K w_k g_{jk}\right) - \sum_{k=1}^K w_k S^r(g_{jk}) \right| \quad (6)$$

The relation (6) shows that the weighting factors w_k , $k = 1, \dots, K$, determined to predict the location u_c , reflect the similarity of diffusion properties and can reduce the difference between $S^r(u_c)$ and $\sum_{k=1}^K w_k S^r(g_{jk})$ depending on the local diffusion patterns similarity.

The second term, $\sum_{k=1}^K w_k (S^r(g_{jk}) - S(g_{jk}))$, represents the error term from the measured noises. For the second error term, since the noise variance of the synthesized diffusion MR signal at u_c is $\text{Var}(\sum_{k=1}^K w_k (S^r(g_{jk}) - S(g_{jk}))) = \sum_{k=1}^K w_k^2 \text{Var}(S^r(g_{jk}) - S(g_{jk}))$, the

noise level of the synthesized diffusion MR signal at the predicted location is less than the noise level of measured diffusion MR signals at each DG direction, assuming that the noise is independent and identically distributed.

2.4. High resolution MR diffusion signals

For a given voxel coordinate r and measured MR diffusion signals $S(g_i)$, $i = 1, \dots, N$, we aim to generate appropriate diffusion MR signals on a finer set $\mathcal{D}_M = \{g_1, \dots, g_N, v_1, \dots, v_{M-N}\}$. Let $\mathcal{D}_C = \{g_1, \dots, g_N, u_{c1}, \dots, u_{cL}\}$ be the total set of measured diffusion gradient directions g_i and the synthesized directions u_{cL} estimated by the diffusion pattern similarity in (4). We use the radial basis function (RBF) approximation to determine the MR signals $S(v_j)$, $j = 1, \dots, M - N$. The MR diffusion signals $S(v_j)$ are defined by

$$S(v_j) = \sum_{g \in \mathcal{D}_C} c_g \phi(v_j - g), \quad j = 1, \dots, M - N, \quad (7)$$

where ϕ is a positive definite radial basis function and c_g , $g \in \mathcal{D}_C$, are the weight coefficients. In this paper, we use Gaussian function $\phi(\cdot) = e^{-\alpha \|\cdot\|^2}$. To estimate the weight coefficients c_g , we solve the following linear system,

$$(A + \lambda I)c = b \quad (8)$$

where

$$A = \begin{bmatrix} e^{-\alpha \|g_1 - g_1\|^2} & \dots & e^{-\alpha \|g_1 - u_{cL}\|^2} \\ \vdots & \ddots & \vdots \\ e^{-\alpha \|u_{cL} - g_1\|^2} & \dots & e^{-\alpha \|u_{cL} - u_{cL}\|^2} \end{bmatrix},$$

$$c = \begin{bmatrix} c_1 \\ \vdots \\ c_{N+L} \end{bmatrix}, \text{ and } b = \begin{bmatrix} S(g_1) \\ \vdots \\ S(u_{cL}) \end{bmatrix}.$$

Here, λ is a regularization parameter and I denotes the identity matrix. The resulting weight coefficients vector c are the solution of the following regularized least squares approximation problem ((Fasshauer, 2007)):

$$\min_{c \in \mathbb{R}^{N+L}} \left\{ (Ac - b)^T (Ac - b) + \frac{\lambda}{2} c^T A c \right\} \quad (9)$$

The regularization parameter λ controls the balance between the smoothness of interpolation function and the data fitting. The implementation procedure of the proposed algorithm is summarized as follows: